

CS 161A: Programming and Problem Solving I

Assignment 4: Phrases and subphrases



Academic Integrity

You may NOT, under any circumstances, begin a programming assignment by looking for completed code on StackOverflow or Chegg or any such website, which you can claim as your own. Please check out the [Student Code of Conduct at PCC](#).

The only way to learn to code is to do it yourself. The assignments will be built from examples during the lectures, so ask for clarification during class if something seems confusing. If you start with code from another source and just change the variable names or other content to make it look original, you will receive a zero on the assignment.

I may ask you to explain your assignment verbally. If you cannot satisfactorily explain what your code does, and answer questions about why you wrote it in a particular way, then you should also expect a zero.

Introduction

A phrase is a small group of words that communicates a concept but isn't a full sentence. You use phrases in your writing and your speech every day. There are lots of different kinds of phrases, some of which play a technical role in your writing and others that play a more illustrative role. No matter which role a phrase is playing, it's achieving one simple goal: making your sentences richer by giving your words context, detail, and clarity.



Purpose

In the last assignment you used branch statements, or if/else statements, to choose which code to execute depending on the result of a conditional expression using numbers. In this assignment you will continue to write conditional expressions using if statements for comparing strings of text.

After completing this assignment you will be able to:

- Use if/else statements to perform different string operations on a string input
- Use the [find\(\)](#) function, [substr](#) function, and [string::npos](#) constant to determine whether a particular string is in another string
- Include the `<string>` header file in your program to access string functions

Task

- ☐ Before you get started:
 - ☐ Check out Sample Assignment A04 - [Algorithmic Design document](#)
 - ☐ Check out Sample Assignment A04 - [Sample4 Code](#)
- ☐ Open the [Algorithmic Design Document](#), make a copy, and follow the steps to create your algorithm.
- ☐ You must express your algorithm as **pseudocode** or a **flowchart**. You may use generative AI to help you complete the algorithm, but you must write your own code.
- ☐ Your task is to write a program that reads two phrases on separate lines and outputs one of the four responses: (See Sample Run below in [Criteria for Success](#))
 - Phrase one is found within phrase two
 - Phrase two is found within phrase one
 - Both phrases match
 - No matches
- ☐ Print a welcome message for your program.
- ☐ Prompt the user to enter the first phrase. Store in a **string** variable.
- ☐ Echo the phrase entered back to the screen.
- ☐ Prompt the user to enter the second phrase. Store in a **string** variable.
- ☐ Echo the phrase entered back to the screen.
- ☐ You must use the [find\(\)](#) function, [substr](#) function and [string::npos](#) constant to do the following:
 - ☐ If the phrase 1 is found in phrase 2, then print Phrase 1 is found within phrase 2, and also print phrase 2 from the position where phrase 1 is found.
 - ☐ If the phrase 2 is found in phrase 1, then print Phrase 2 is found within phrase 1, and also print phrase 1 from the position where phrase 2 is found.

- ☐ If the phrases match, then print Both phrases match.
- ☐ If the phrases don't match, then print No match.
- ☐ Print a Thank You message to end your program.

Criteria for Success

- ☐ Test your program using the following sample runs, making sure you get the same output when using the given inputs (in **blue**):

```
Welcome to the Phrases and Subphrases program!
```

```
Enter Phrase 1: truck
```

```
You entered: truck
```

```
Enter Phrase 2: firetruck is here
```

```
You entered: firetruck is here
```

```
truck is found in firetruck is here  
truck is here
```

```
Thank you for using my program!
```

```
Welcome to the Phrases and Subphrases program!
```

```
Enter Phrase 1: the green grass grows
```

```
You entered: the green grass grows
```

```
Enter Phrase 2: green grass
```

```
You entered: green grass
```

```
green grass is found in the green grass grows  
green grass grows
```

```
Thank you for using my program!
```

```
Welcome to the Phrases and Subphrases program!
```

```
Enter Phrase 1: He was between a rock and a hard place
```

```
You entered: He was between a rock and a hard place
```

```
Enter Phrase 2: rock
```

```
You entered: rock
```

```
rock is found in He was between a rock and a hard place  
rock and a hard place
```

```
Thank you for using my program!
```

- ☐ Check out Sample Assignment A04 - [Algorithmic Design document](#)
- ☐ Check out Sample Assignment A04 - [Sample4 Code](#)
- ☐ Complete zyBooks section **CS161A 4. Conditionals Part II** activities.

- ☐ Complete all sections of your Algorithmic Design Document.
- ☐ Include **pseudocode** in part d of the design document. You may use generative AI to help you complete the algorithm, but you must write your own code.
- ☐ **Follow these Coding Construct Requirements:**
 - ☐ Must have all the right data types mentioned under Task.
 - ☐ Must use all the appropriate string functions.
 - ☐ Read the 2 phrases on separate lines as shown in the Sample Run.
 - ☐ Print a welcome and goodbye message.
- ☐ Please open and compare your work with the [grading rubric](#) before submitting.
- ☐ Remember to follow all [style guidelines](#).
- ☐ You must have comments in your code per [style guidelines](#).
- ☐ Download your Algorithmic Design Document as a PDF (File -> Download -> PDF), rename it to `a04.pdf`, and upload it to the D2L assignment by **Wednesday**.
- ☐ Upload your `a04.cpp` C++ source file to the D2L assignment by **Sunday**.
- ☐ Do your own work. Consult the syllabus for more information about academic integrity.

Additional Support

- ☐ Post a question for the instructor in the Ask Questions! area of the Course Lobby.
- ☐ Check out Sample Assignment A04 - [Algorithmic Design document](#)
- ☐ Check out Sample Assignment A04 - [Sample4 Code](#)
- ☐ [The string .find function reference](#)
- ☐ [The string .substr\(\) function reference](#).